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Tecnológico de Estudios Superiores de Villa Guerrero

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Carrera: Ingeniería en Sistemas Computacionales

Nombre de la Materia: Sistemas Operativos

Practica_3: Explorando el hardware de tu Sistema con lspci

Grupo:3501

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INTRODUCCION

En los sistemas Linux, la gestión del hardware depende del bus PCI (Peripheral Component Interconnect) y de los módulos del kernel (drivers) que permiten la interacción con componentes como la tarjeta gráfica o la de red. En esta práctica se utiliza el comando `lspci` para identificar estos dispositivos físicos y verificar los controladores que los gestionan.



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1. Ver que hay en tu sistema.

COMANDO: lspci.

¿Qué hace? Muestra una lista resumida de todos los dispositivos conectados al bus PCI de la computadora.

```
user@debian:~$ lspci
0000:00:00.0 Host bridge: Intel Corporation Tiger Lake-UP3/W35 4 cores Host Bridge/DRAM Registers (rev 02)
0000:00:02.0 VGA compatible controller: Intel Corporation TigerLake-LP GT2 (Iris Xe Graphics) (rev 03)
0000:00:04.0 Signal processing controller: Intel Corporation TigerLake-LP Dynamic Tuning Processor Participant (rev 02)
0000:00:0a.0 Signal processing controller: Intel Corporation TigerLake Telemetry Aggregator Driver (rev 01)
0000:00:0e.0 USB controller: Intel Corporation Tiger Lake-LP Thunderbolt 4 USB Controller (rev 02)
0000:00:0e.0 RAID bus controller: Intel Corporation Volume Management Device NVMe RAID Controller
0000:00:14.0 USB controller: Intel Corporation Tiger Lake-LP USB 3.2 Gen 2x1 xHCI Host Controller (rev 30)
0000:00:14.2 RAM memory: Intel Corporation Tiger Lake-LP Shared SRAM (rev 30)
0000:00:14.3 Network controller: Intel Corporation Wi-Fi 6 AX201 (rev 30)
0000:00:15.0 Serial bus controller: Intel Corporation Tiger Lake-LP Serial IO I2C Controller #0 (rev 30)
0000:00:15.1 Serial bus controller: Intel Corporation Tiger Lake-LP Serial IO I2C Controller #1 (rev 30)
0000:00:15.6 Communication controller: Intel Corporation Tiger Lake-LP Management Engine Interface (rev 30)
0000:00:17.0 SATA controller: Intel Corporation Tiger Lake-LP SATA Controller (rev 30)
0000:00:1a.0 USB controller: Intel Corporation Tiger Lake-LP Serial IO UHCI Controller #0 (rev 30)
0000:00:1a.3 Serial bus controller: Intel Corporation Tiger Lake-LP Serial IO SPI Controller #1 (rev 30)
0000:00:1f.0 ISA bridge: Intel Corporation Tiger Lake-LP LPC Controller (rev 30)
0000:00:1f.3 Multimedia audio controller: Intel Corporation Tiger Lake-LP Smart Sound Technology Audio Controller (rev 30)
0000:00:1f.4 SMBus: Intel Corporation Tiger Lake-LP SMBus Controller (rev 30)
0000:00:1f.5 Serial bus controller: Intel Corporation Tiger Lake-LP SPI Controller (rev 30)
1000:e9:1c:0 System peripheral: Intel Corporation RST VMD Managed Controller
1000:e9:1c:4 PCI bridge: Intel Corporation Tiger Lake-LP PCI Express Root Port #5 (rev 30)
1000:e1:00:0 Non-Volatile memory controller: SK hynix BC981 NVMe Solid State Drive (DRAM-less) (rev 03)
user@debian:~$
```

2. Controlador.

COMANDO: lspci -v

¿Qué hace? Muestra información detallada e inspección avanzada de cada dispositivo PCI.

```
user@debian:~$ lspci -v
0000:00:00.0 Host bridge: Intel Corporation Tiger Lake-UP3/W35 4 cores Host Bridge/DRAM Registers (rev 02)
    Flags: bus master, fast devsel, latency 0, IOMMU group 1
    Capabilities: <access denied>
    Kernel modules: igbm, edac

0000:00:02.0 VGA compatible controller: Intel Corporation TigerLake-LP GT2 (Iris Xe Graphics) (rev 03) (prog-if 00 [VGA controller])
    Subsystem: Lenovo Device 379c
    Flags: bus master, fast devsel, latency 0, IRQ 359, IOMMU group 0
    Memory at 6002000000 (64-bit, non-prefetchable) [size=16M]
    Memory at 4000000000 (64-bit, prefetchable) [size=256M]
    I/O ports at 3000 [size=64]
    Expansion ROM at 400c0000 [virtual] [disabled] [size=128K]
    Capabilities: <access denied>
    Kernel driver in use: i915
    Kernel modules: i915, xe

0000:00:04.0 Signal processing controller: Intel Corporation TigerLake-LP Dynamic Tuning Processor Participant (rev 02)
    Subsystem: Lenovo Device 380a
    Flags: bus master, fast devsel, latency 0, IRQ 36, IOMMU group 2
    Memory at 6003200000 (64-bit, non-prefetchable) [size=128K]
    Capabilities: <access denied>
    Kernel driver in use: proc, thermal
    Kernel modules: processor_thermal_device_pci_legacy

0000:00:0a.0 Signal processing controller: Intel Corporation TigerLake Telemetry Aggregator Driver (rev 01)
    Subsystem: Lenovo Device 3801
    Flags: fast devsel, IOMMU group 3
    Memory at 6003200000 (64-bit, non-prefetchable) [size=32K]
    Capabilities: <access denied>
    Kernel driver in use: intel_vsec
    Kernel modules: intel_vsec

0000:00:0e.0 USB controller: Intel Corporation Tiger Lake-LP Thunderbolt 4 USB Controller (rev 02) (prog-if 30 [XHCI])
    Subsystem: Lenovo Device 3802
    Flags: medium devsel, IRQ 143, IOMMU group 4
    Memory at 6003200000 (64-bit, non-prefetchable) [size=64K]
    Capabilities: <access denied>
    Kernel driver in use: xhci_hcd
    Kernel modules: xhci_pci

0000:00:0e.0 RAID bus controller: Intel Corporation Volume Management Device NVMe RAID Controller (prog-if 00 [vendor specific])
    Subsystem: Intel Corporation Device 7270
    Flags: bus master, fast devsel, latency 0, IOMMU group 5
    Memory at 6000000000 (64-bit, non-prefetchable) [size=32M]
    Memory at 520000000 (32-bit, non-prefetchable) [size=2M]
    Memory at 6003100000 (64-bit, non-prefetchable) [size=3M]
    Capabilities: <access denied>
    Kernel driver in use: vmd
    Kernel modules: vmd

0000:00:14.0 USB controller: Intel Corporation Tiger Lake-LP USB 3.2 Gen 2x1 xHCI Host Controller (rev 30) (prog-if 30 [XHCI])
    Subsystem: Lenovo Device 3817
    Flags: bus master, medium devsel, latency 0, IRQ 151, IOMMU group 6
    Memory at 54000000 (64-bit, non-prefetchable) [size=64K]
    Capabilities: <access denied>
    Kernel driver in use: xhci_hcd
    Kernel modules: xhci_pci

0000:00:14.2 RAM memory: Intel Corporation Tiger Lake-LP Shared SRAM (rev 30)
    Subsystem: Lenovo Device 3816
    Flags: bus master, fast devsel, latency 0, IOMMU group 6
    Memory at 6002400000 (64-bit, non-prefetchable) [size=4K]
    Memory at 6002400000 (64-bit, non-prefetchable) [size=4K]
    Capabilities: <access denied>

0000:00:14.3 Network controller: Intel Corporation Wi-Fi 6 AX201 (rev 30)
    Subsystem: Intel Corporation Device 0074
    Flags: bus master, fast devsel, latency 0, IRQ 30, IOMMU group 7
    Memory at 6003200000 (64-bit, non-prefetchable) [size=16K]
    Capabilities: <access denied>
    Kernel driver in use: iwlwifi
    Kernel modules: iwlwifi

0000:00:15.0 Serial bus controller: Intel Corporation Tiger Lake-LP Serial IO I2C Controller #0 (rev 30)
    Subsystem: Lenovo Device 3813
    Flags: bus master, fast devsel, latency 0, IRQ 27, IOMMU group 8
    Memory at 4017000000 (64-bit, non-prefetchable) [size=4K]
    Capabilities: <access denied>
    Kernel driver in use: intel_lpss
    Kernel modules: intel_lpss_pci

0000:00:15.1 Serial bus controller: Intel Corporation Tiger Lake-LP Serial IO I2C Controller #1 (rev 30)
    Subsystem: Lenovo Device 3812
```



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```
Oct 3 6:12 PM
user@debian:~$ lsusb
0000:00:15.1 Serial bus controller: Intel Corporation Tiger Lake-LP Serial IO I2C Controller #1 (rev 30)
Subsystem: Lenovo Device 3812
Flags: bus master, fast devsel, latency 0, IRQ 40, IOMMU group 8
Memory at 4017001000 (64-bit, non-prefetchable) [size=4K]
Capabilities: <access denied>
Kernel driver in use: intel-lpsx
Kernel modules: intel-lpsx_pci

0000:00:16.0 Communication controller: Intel Corporation Tiger Lake-LP Management Engine Interface (rev 30)
Subsystem: Lenovo Device 380f
Flags: bus master, fast devsel, latency 0, IRQ 170, IOMMU group 9
Memory at 6003247000 (64-bit, non-prefetchable) [size=4K]
Capabilities: <access denied>
Kernel driver in use: mei_me
Kernel modules: mei_me

0000:00:17.0 SATA controller: Intel Corporation Tiger Lake-LP SATA Controller (rev 30) (prog-if 01 [AHCI 1.0])
Subsystem: Intel Corporation Device 7770
Flags: bus master, 66MHz, medium devsel, latency 0, IRQ 168, IOMMU group 10
Memory at 54010000 (32-bit, non-prefetchable) [size=4K]
Memory at 54014000 (32-bit, non-prefetchable) [size=256]
I/O ports at 3000 [size=4]
I/O ports at 3008 [size=32]
Memory at 54013000 (32-bit, non-prefetchable) [size=2K]
Capabilities: <access denied>
Kernel driver in use: ahci
Kernel modules: ahci

0000:00:1e.0 Communication controller: Intel Corporation Tiger Lake-LP Serial IO UART Controller #0 (rev 30)
Subsystem: Lenovo Device 380a
Flags: bus master, fast devsel, latency 0, IRQ 10, IOMMU group 11
Memory at 4017002000 (64-bit, non-prefetchable) [size=4K]
Capabilities: <access denied>
Kernel driver in use: intel-lpsx
Kernel modules: intel-lpsx_pci

0000:00:1e.3 Serial bus controller: Intel Corporation Tiger Lake-LP Serial IO SPI Controller #1 (rev 30)
Subsystem: Lenovo Device 3810
Flags: bus master, fast devsel, latency 0, IRQ 37, IOMMU group 11
Memory at 4017003000 (64-bit, non-prefetchable) [size=4K]
Capabilities: <access denied>
Kernel driver in use: intel-lpsx

Oct 3 6:13 PM
user@debian:~$ lsusb
0000:00:1f.0 ISA bridge: Intel Corporation Tiger Lake-LP LPC Controller (rev 30)
Subsystem: Lenovo Device 3803
Flags: bus master, fast devsel, latency 0, IOMMU group 12

0000:00:1f.3 Multimedia audio controller: Intel Corporation Tiger Lake-LP Smart Sound Technology Audio Controller (rev 30)
Subsystem: Lenovo Device 380a
Flags: bus master, fast devsel, latency 32, IRQ 181, IOMMU group 12
Memory at 6003200000 (64-bit, non-prefetchable) [size=16K]
Memory at 6003000000 (64-bit, non-prefetchable) [size=1M]
Capabilities: <access denied>
Kernel driver in use: sof-usb-pci-intel-tgl
Kernel modules: snd_hda_intel, snd_soc_aws, snd_soc_pci_intel_tgl

0000:00:1f.4 SMBus: Intel Corporation Tiger Lake-LP SMBus Controller (rev 30)
Subsystem: Lenovo Device 3800
Flags: medium devsel, IRQ 16, IOMMU group 12
Memory at 6003244000 (64-bit, non-prefetchable) [size=256]
I/O ports at efa0 [size=32]
Kernel driver in use: i801_smbus
Kernel modules: i2c, i801

0000:00:1f.5 Serial bus controller: Intel Corporation Tiger Lake-LP SPI Controller (rev 30)
Subsystem: Lenovo Device 3808
Flags: bus master, fast devsel, latency 0, IOMMU group 12
Memory at 50400000 (32-bit, non-prefetchable) [size=4K]
Kernel modules: i2c, i801

10000:e0:1c.0 System peripheral: Intel Corporation HST VMD Managed Controller
Flags: fast devsel, IOMMU group 5

10000:e0:1c.4 PCI bridge: Intel Corporation Tiger Lake-LP PCI Express Root Port #5 (rev 30) (prog-if 00 [Normal decode])
Subsystem: Intel Corporation Device 7770
Flags: bus master, fast devsel, latency 0, IRQ 141, IOMMU group 5
Bus: primary00, secondary0, subordinate0e1, sec-latency0
I/O behind bridge: [disabled] [16-bit]
Memory behind bridge: 52000000-52000000 [size=1M] [32-bit]
Prefetchable memory behind bridge: [disabled] [64-bit]
Capabilities: <access denied>
Kernel driver in use: pcieport

10000:e1:00.0 Non-Volatile Memory controller: SK hynix BC901 NVMe Solid State Drive (DRAM-less) (rev 03) (prog-if 02 [NVM Express])
Subsystem: SK hynix BC901 NVMe Solid State Drive (DRAM-less)
Flags: bus master, fast devsel, latency 0, IRQ 2147483648, IOMMU group 5
Memory at 52000000 (64-bit, non-prefetchable) [size=16K]
Kernel driver in use: nvme
```

PREGUNTAS:

¿Qué driver está usando tu tarjeta de red Wi-Fi (Wireless)?

Dispositivo: Intel Corporation Wi-Fi 6 AX201

Driver en uso: iwlwifi

```
0000:00:14.3 Network controller: Intel Corporation Wi-Fi 6 AX201 (rev 30)
Subsystem: Intel Corporation Device 0874
Flags: bus master, fast devsel, latency 0, IRQ 16, IOMMU group 7
Memory at 600323c000 (64-bit, non-prefetchable) [size=16K]
Capabilities: <access denied>
Kernel driver in use: iwlwifi
Kernel modules: iwlwifi
```



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¿Qué driver está usando tu tarjeta gráfica?

Dispositivo: Intel Corporation TigerLake-LP GT2 [Iris Xe Graphics]

Driver en uso: i915

```
0000:80:02.0 VGA compatible controller: Intel Corporation TigerLake-LP GT2 [Iris Xe Graphics] (rev 03) (prog-if 00 [VGA controller])
Subsystem: Lenovo Device 3f9c
Flags: bus master, fast devsel, latency 0, IRQ 169, IOMMU group 0
Memory at 6002000000 (64-bit, non-prefetchable) [size=16M]
Memory at 4000000000 (64-bit, prefetchable) [size=256M]
I/O ports at 3900 [size=64]
Expansion ROM at 000c0000 [virtual] [disabled] [size=128K]
Capabilities: <access denied>
Kernel driver in use: i915
Kernel modules: i915, xe
```



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CONCLUSION

La ejecución de la práctica permitió comprender el uso de la herramienta lspci para la inspección del bus PCI en sistemas Linux, facilitando la identificación de los componentes de hardware y la verificación de los controladores activos en el kernel (iwlwifi para la tarjeta de red Wi-Fi e i915 para la tarjeta gráfica). Esto demuestra la importancia de la interfaz de línea de comandos en la administración y diagnóstico de los dispositivos del sistema.

